UNIT 10

ELECTRONICS AND COMMUNICATION

Electronics is clearly the winner of the day
– John Ford.

Learning Objectives

In this unit, the students are exposed to

- Energy band diagram in semiconductors
- Types of semiconductors
- Formation of *p-n* junction diode and its V-I characteristics
- Rectification process
- Special purpose diodes
- Transistors and their immediate applications
- Digital and analog signals
- Logic gates, Boolean algebra and De Morgan's theorem
- Modulation and its types
- Basic elements of communication system
- Propagation of electromagnetic waves through space
- Some important communication systems

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10.1 INTRODUCTION

Electronics has become a part of our daily life. All gadgets like mobile phones, computers, televisions, music systems etc work on the electronic principles. Electronic circuits are used to perform various operations in devices like air conditioners, microwave oven, dish washers and washing machines. Besides this, its applications are widespread in all fields like communication systems, medical diagnosis and treatments and even handling money through ATMs.

Evolution of Electronics:

The history of electronics began with the invention of vacuum diode by J.A. Fleming in 1897. This was followed by a vacuum triode implemented by Lee De Forest to control electrical signals. This led to the introduction of tetrode and pentode tubes.

Subsequently, the transistor era began with the invention of bipolar junction transistor by Bardeen, Brattain and Shockley in 1948 for which they received Nobel prize in 1956. The emergence of germanium and silicon semiconductor materials made this transistor gain popularity, in turn its application in different electronic circuits.

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